

Semantic Selection Control for Structure-Compatible Generalization

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Phase 6: OOD-free frozen-encoder model selection

Abstract

This phase turns semantic retrieval compatibility into an OOD-free model-selection protocol. Candidate retrieval models are grouped into encoder-threshold seed zoos, filtered to high train and ID validation performance, selected by learned compatibility or baseline selectors, and evaluated only afterward on held-out semantic variants.

1. Manifest

Field	Value
GPU	L4
encoders	['all_minilm_l6_v2', 'bge_small_en_v1_5']
thresholds	[0.5, 0.56, 0.62, 0.68, 0.74]
zoos	12
configs per zoo	12
selection records	840

Table 1. Semantic selection Modal L4 manifest.

2. Gate Status

Gate	Passed
min_zoo_count	True
beats_id_validation	True
beats_train_accuracy	True
beats_random_candidate	True
wrong_control_fails	True
accepted	True

Table 2. Pre-registered selection-control gates.

Selector	Zoos	OOD	Regret	Lift	Ties
random_candidate	120	0.919	0.073	0.000	8.617
id_validation_accuracy	120	0.919	0.073	0.000	8.617
train_accuracy	120	0.919	0.073	0.000	8.617
compatibility_wrong	120	0.751	0.242	-0.168	1.967
compatibility_discovered	120	0.978	0.014	0.059	3.917
compatibility_true	120	0.993	0.000	0.073	3.617
ood_oracle	120	0.993	0.000	0.073	3.617

Table 3. Selector performance under train/ID equivalence.

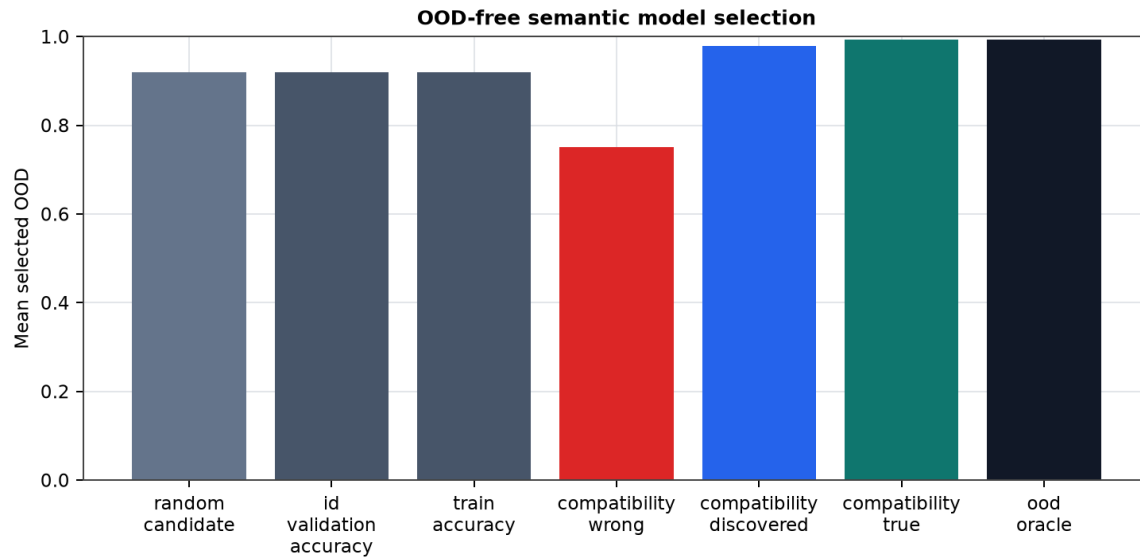


Figure 1. Mean selected OOD by OOD-free selector.

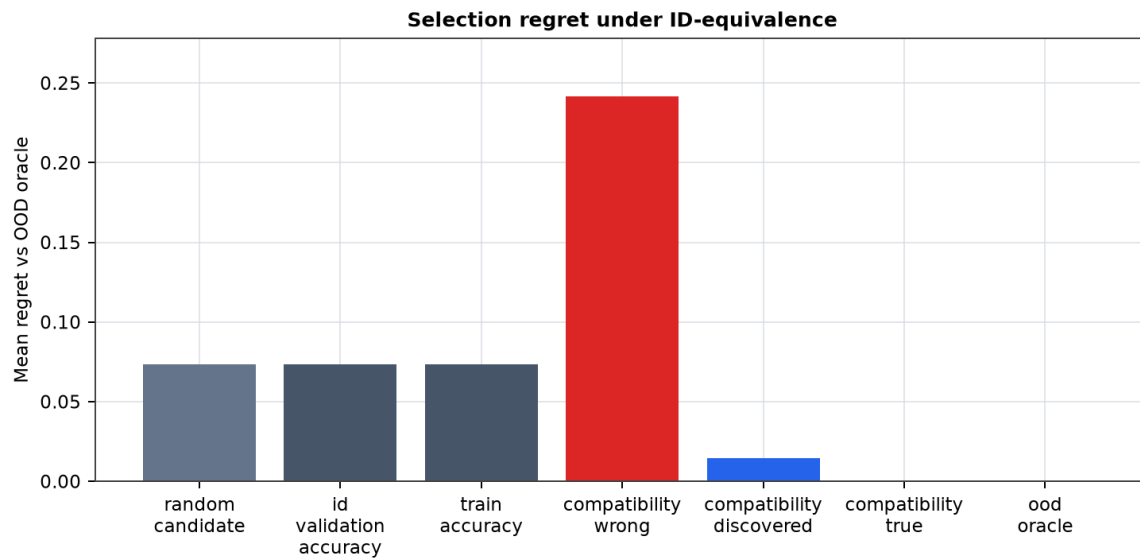


Figure 2. Selection regret versus OOD oracle.

3. Scope

The protocol selects among finite semantic retrieval candidates generated by public frozen encoders. It is an OOD-certifiability-lite result for this structured setting, not a full behavioral guarantee for open-ended language systems.